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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: VII Section: CSD-B

Project Group ID : CSDMP230028

Big Data ML-Based Fake News Detection Using Distributed Learning

Abstract:

The rapid growth of digital media and social platforms has made it increasingly difficult to distinguish authentic news from fake and misleading information. Manual verification is time-consuming and becomes challenging when large volumes of news articles are generated continuously. To address this problem, this project proposes a Big Data ML-Based Fake News Detection system using Distributed Learning for scalable and efficient misinformation detection. The system collects news articles, headlines, source information, and related metadata from datasets, APIs, and multiple sources. Apache Spark is used for distributed data processing, while Natural Language Processing techniques perform data cleaning, duplicate removal, tokenization, text normalization, and feature extraction. TF-IDF, Word2Vec, or transformer-based embeddings are used to convert textual content into machine-readable representations. Machine Learning algorithms such as Logistic Regression, Random Forest, and XGBoost are trained to classify news as Fake, Real, or potentially Misleading. The distributed learning architecture enables processing and model training across multiple workers, improving scalability and supporting large datasets. The system also provides prediction confidence, important features influencing classification, and an analytical dashboard for monitoring fake-news trends, source statistics, and model performance. Evaluation will consider accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, training time, prediction latency, and processing throughput. The project contributes to SDG 16: Peace, Justice and Strong Institutions by supporting access to reliable information and helping address misinformation. It also supports SDG 9: Industry, Innovation and Infrastructure through scalable Big Data and Machine Learning technologies.

Keywords: Big Data, Fake News Detection, Machine Learning, Natural Language Processing, Apache Spark, Distributed Learning, Misinformation Detection, SDG 16, SDG 9

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Mrs. Y. Shakuntala

Signature: _____ Date: _____

Coordinator **Head of the Department**